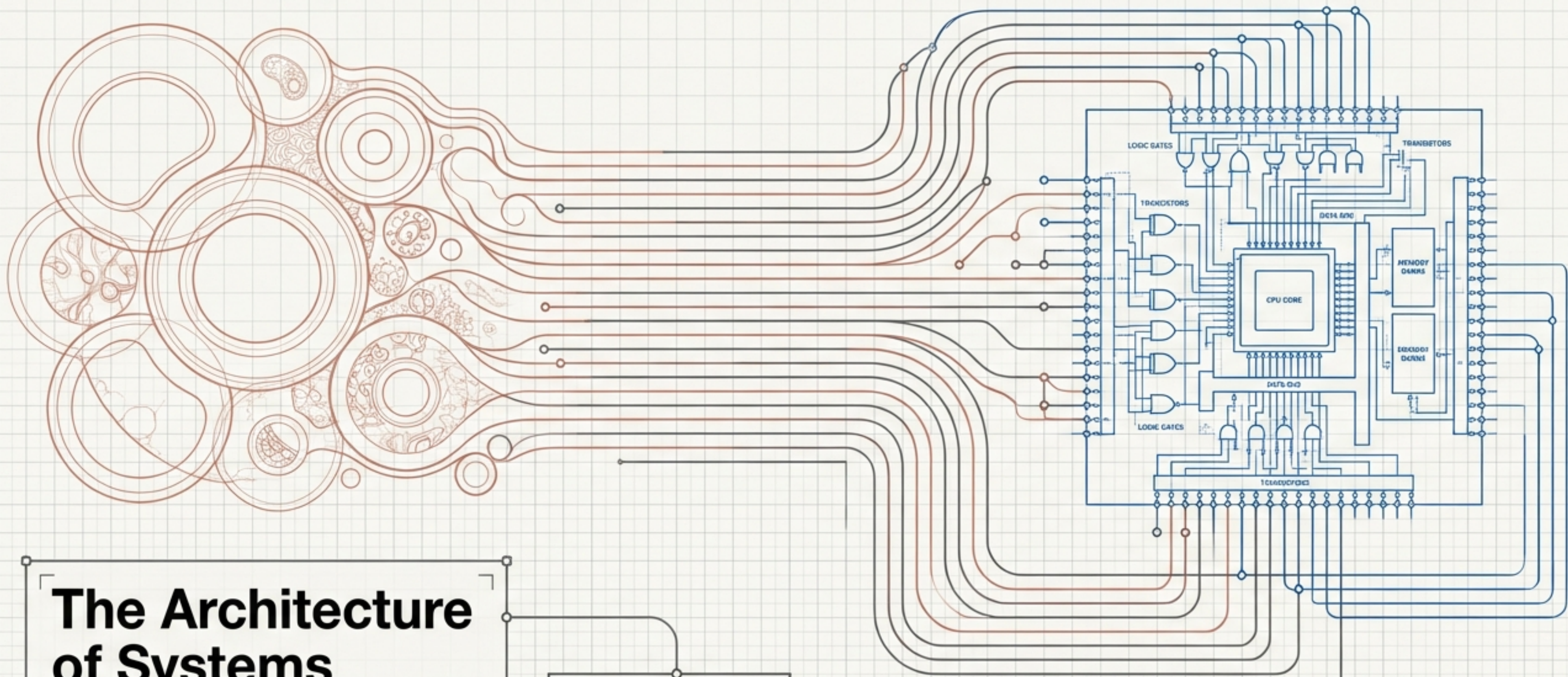




The Architecture of Systems

From universal laws to the microscopic logic of modern computing. A Grade 9 Computer Science introduction.

iCodeGuru
Computer Literacy
Program | Week 01

LEARN, SHARE, AND GROW

THE MISSION

Increasing employability,
introducing high-tech skills, and
connecting students with
international mentors through
100% free courses.

35K+

**COMMUNITY
MEMBERS**

(Up from 5K in 2021)

700+

**INTERNATIONAL
AWARDS**

GLOBAL MILESTONES

Harvard CS50: 259 teams
solved 9/9 puzzles

Stanford Code in Place:
250+ mentors

MIT Emerging Talent:
3 students among 320
selected globally

19

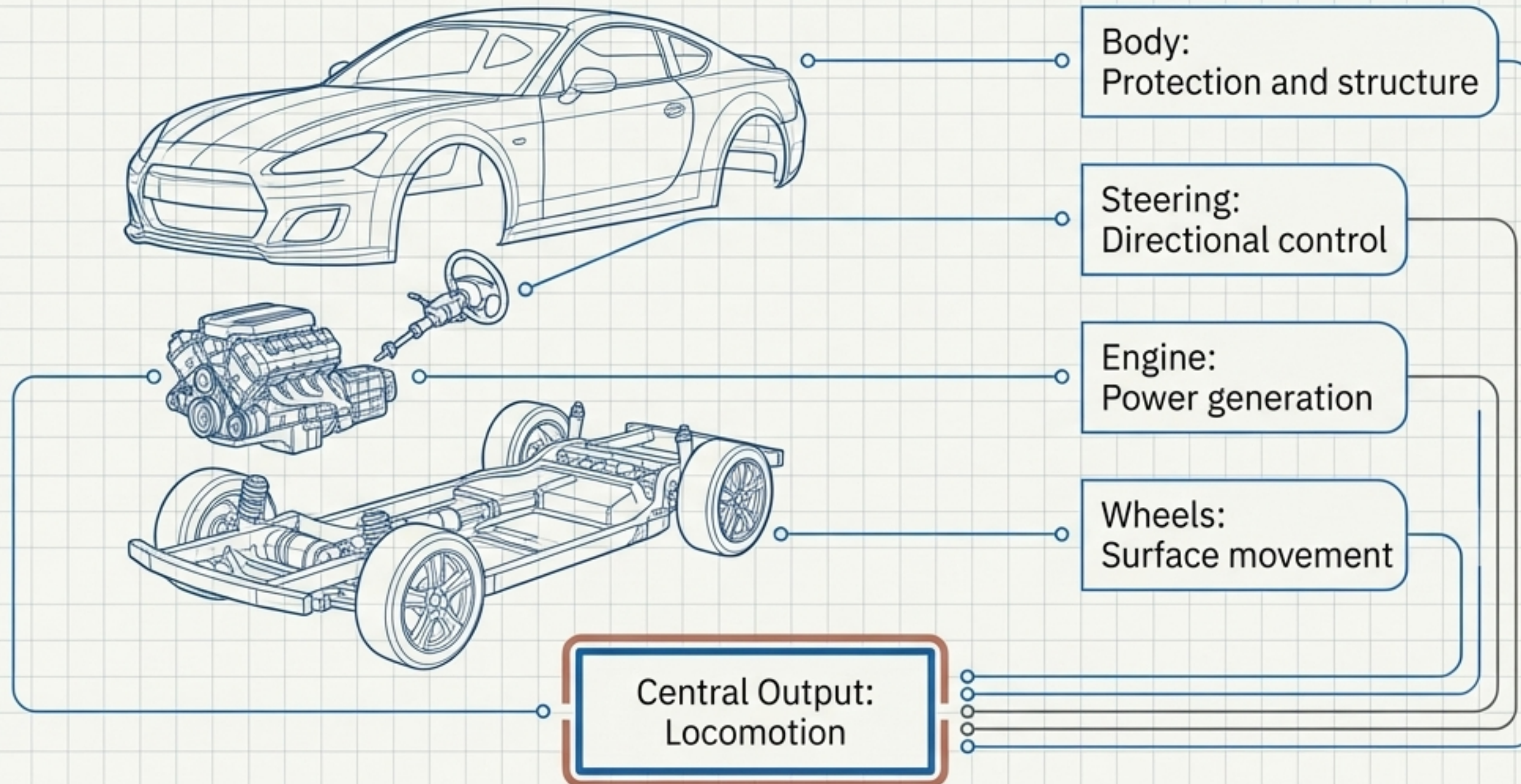
**FULLY FUNDED
USA UNIVERSITIES**

**USA SCHOLARSHIPS
REACHED IN 2024**

Profiles: 3.1-3.8 CGPA,
\$100-\$3000/mo stipends,
zero publications/GRE

DEFINING A SYSTEM

A system is an organized set of components that coordinate to perform a designated function. Every individual component plays a unique role, but the system's true purpose only emerges when they communicate and interrelate.



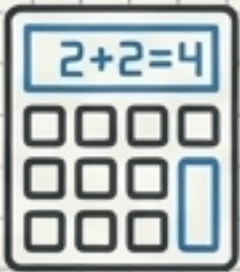
SYSTEM ENVIRONMENTS

STATIC (DETERMINISTIC)

Behavior: Remains unchanged until specific user input is provided.

Output: Highly predictable.

Examples: A basic calculator (waits for the equals button to process $2+2$) or a flashlight (turns on/off via a deliberate switch).



DYNAMIC (NON-DETERMINISTIC)

Behavior: Continuously changes and adapts to surrounding conditions without direct user instruction.

Output: Unpredictable and random.

Examples: Self-driving cars adapting to road conditions, changing traffic signals, and weather forecasting.

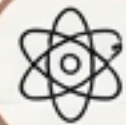


THE TWO WORLDS OF SYSTEMS

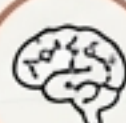
NATURAL SYSTEMS

Operates purely on natural laws without human intervention or modification.

Physical/Chemical: Hydrogen atoms, water molecules.



Biological/Psychological: The human circulatory system, ecosystems, the subconscious brain processing emotions.



ARTIFICIAL SYSTEMS

Designed and organized by humans to solve specific problems.

Engineering: Civil bridges, mechanical robotics, AI like Siri/Alexa.



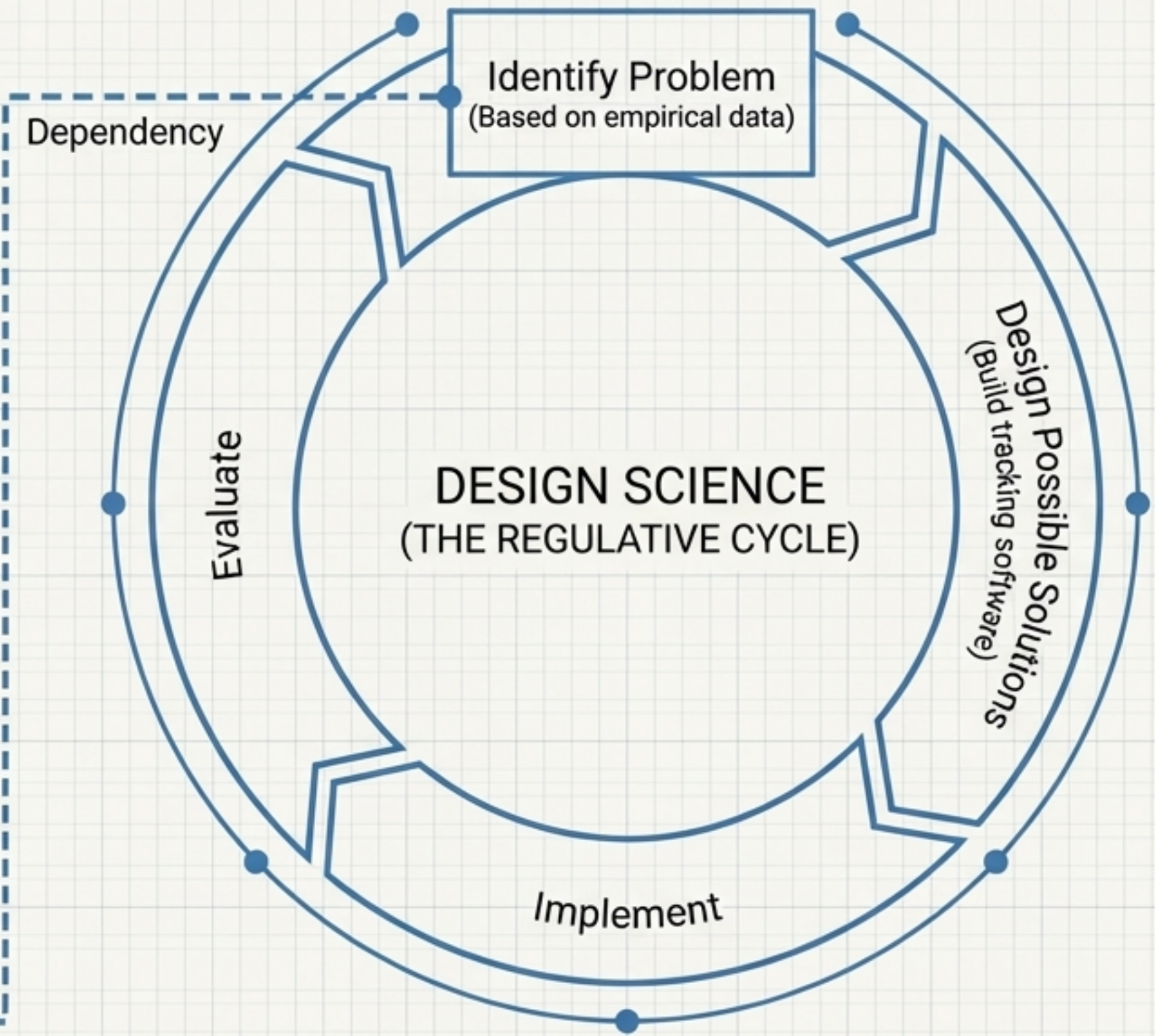
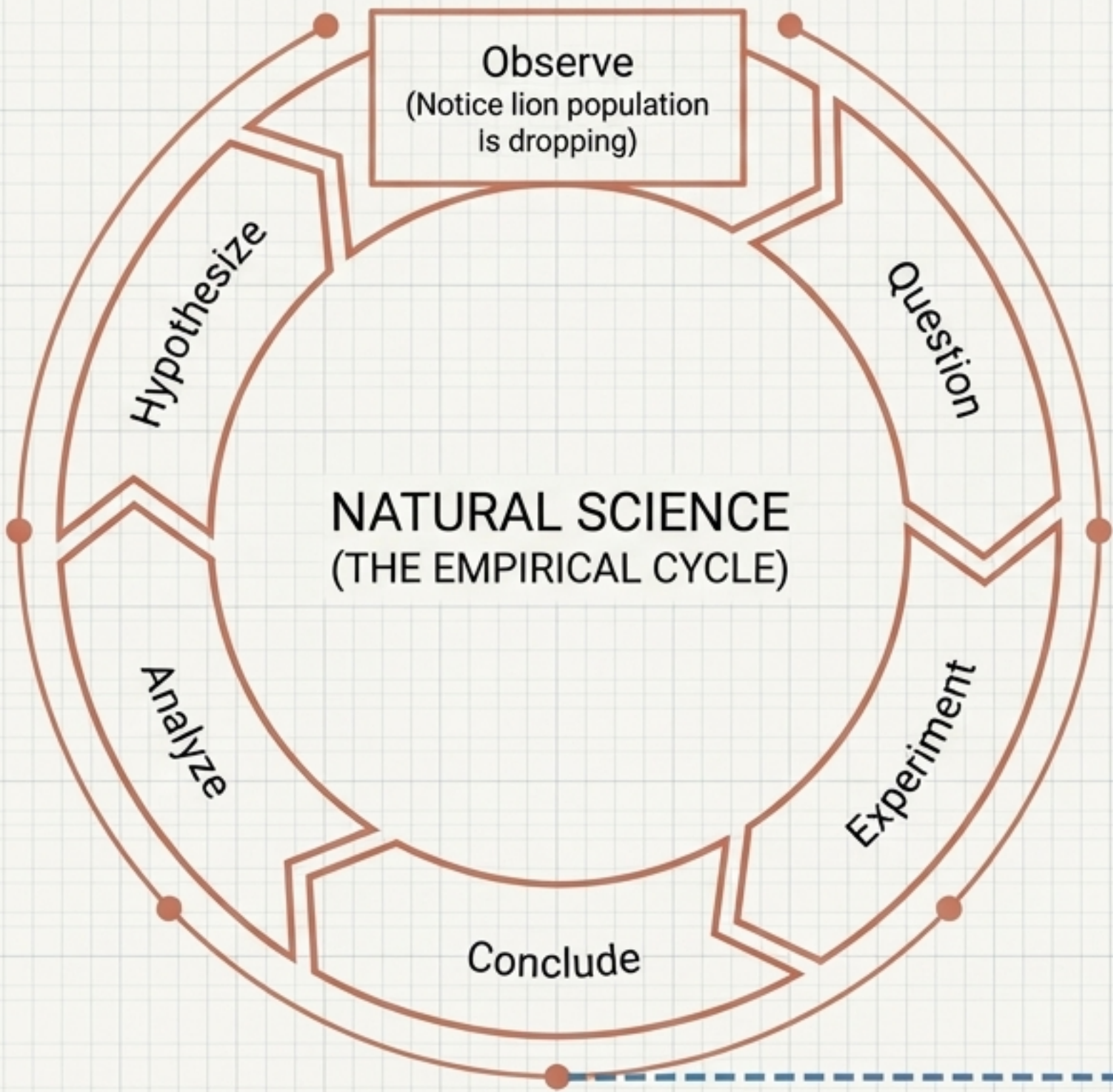
Social: Schools, democracies, the United Nations.



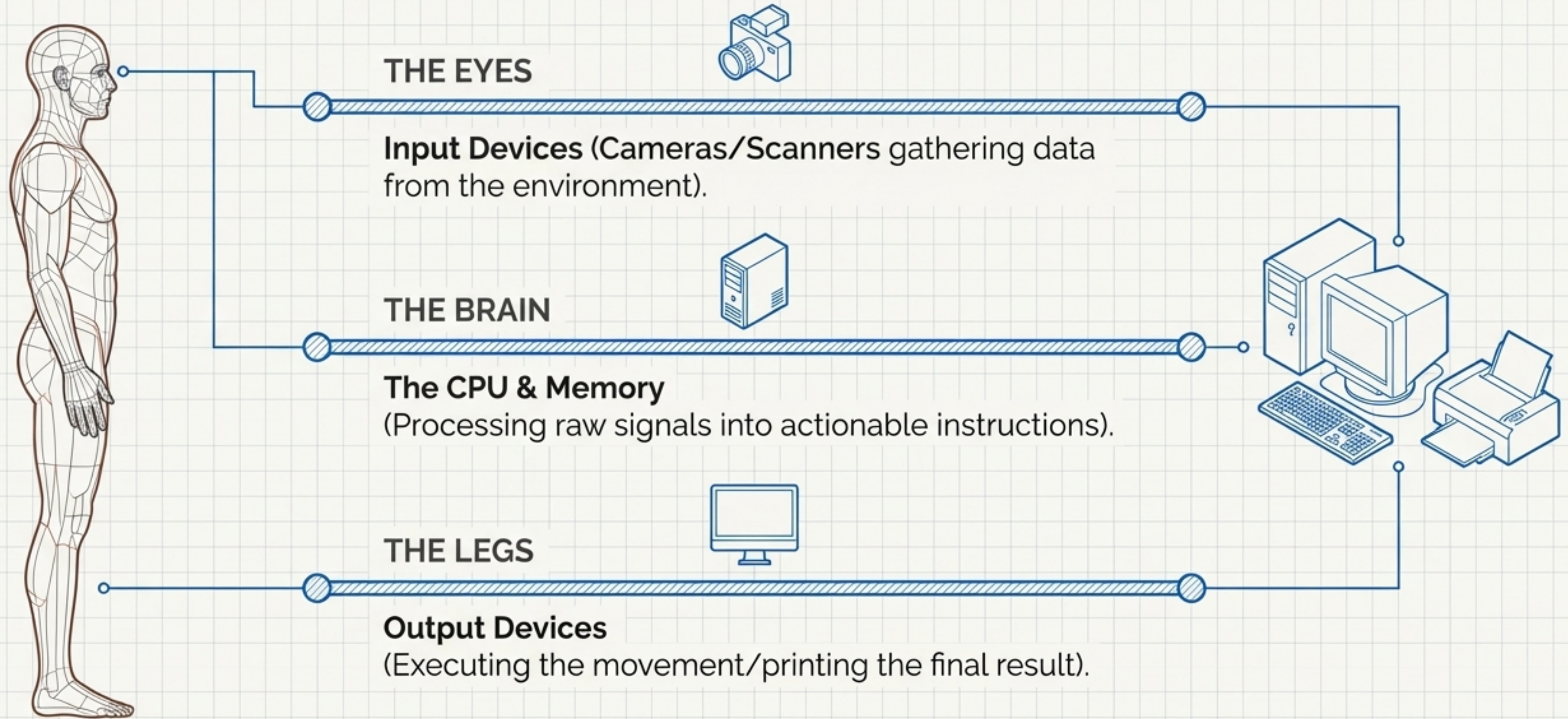
Knowledge: Mathematics, logic, databases (information structured by the human mind).



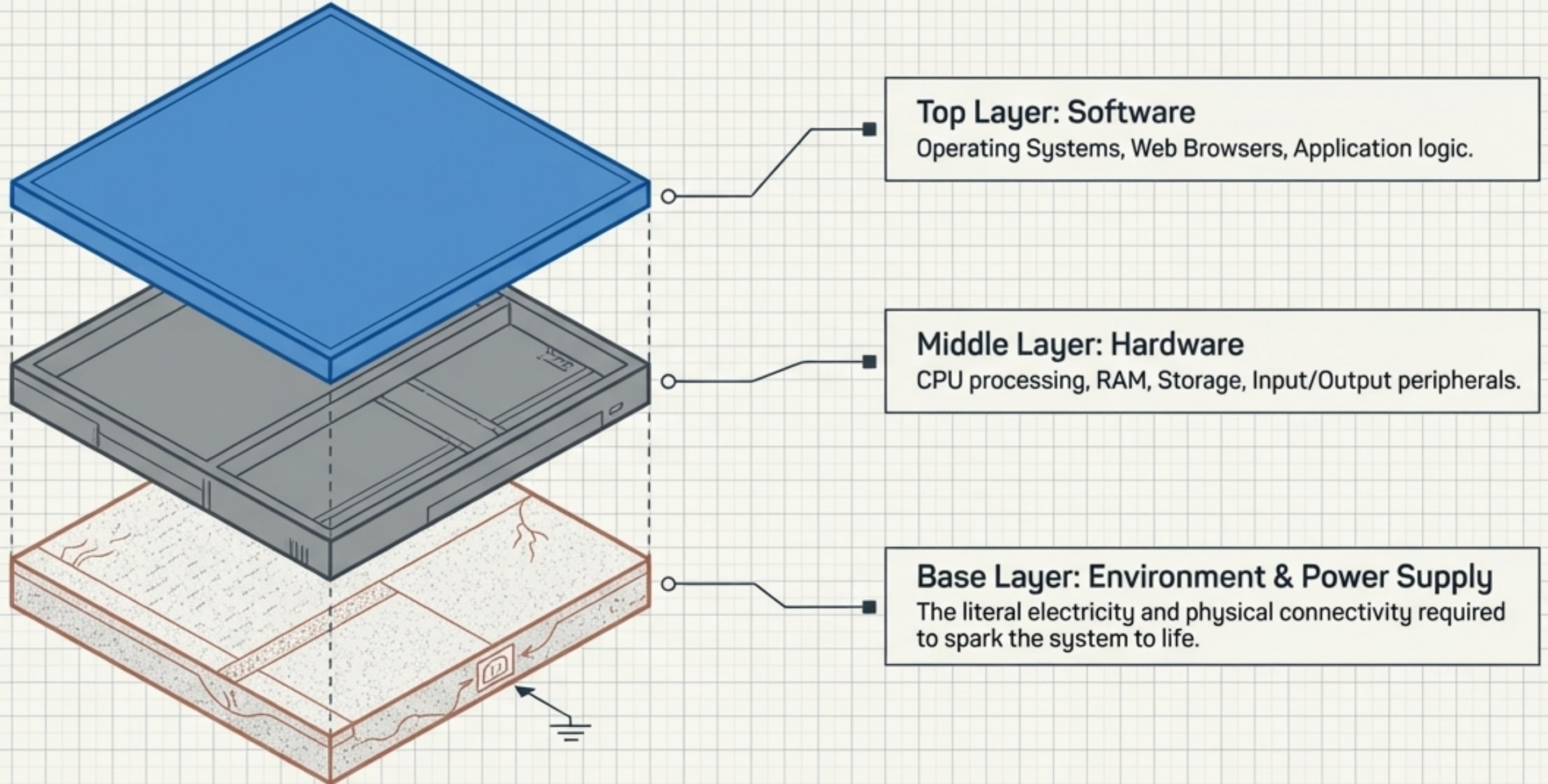
INVESTIGATING THE SYSTEMS



BIOLOGICAL INSPIRATION FOR DIGITAL ARCHITECTURE

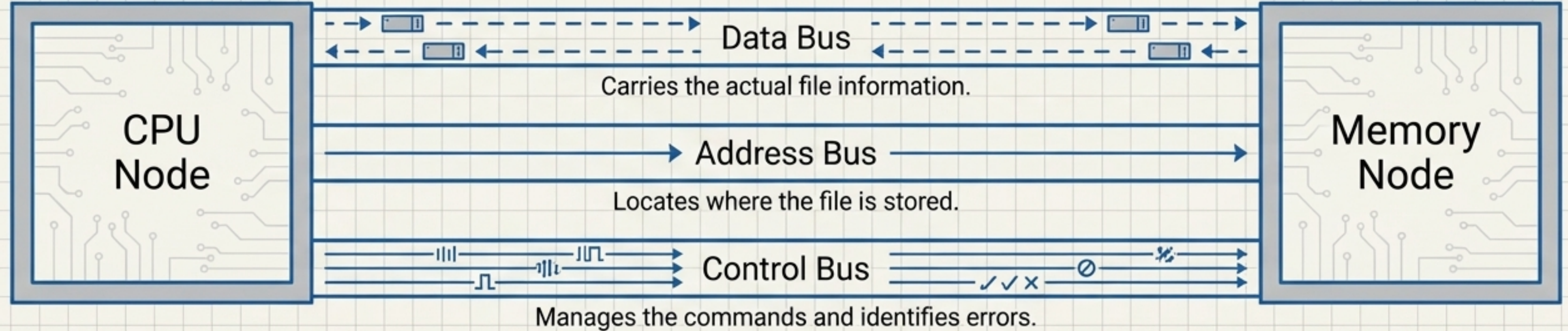


ANATOMY OF A COMPUTING SYSTEM



THE COMMUNICATION PATHWAYS

THE SYSTEM BUS



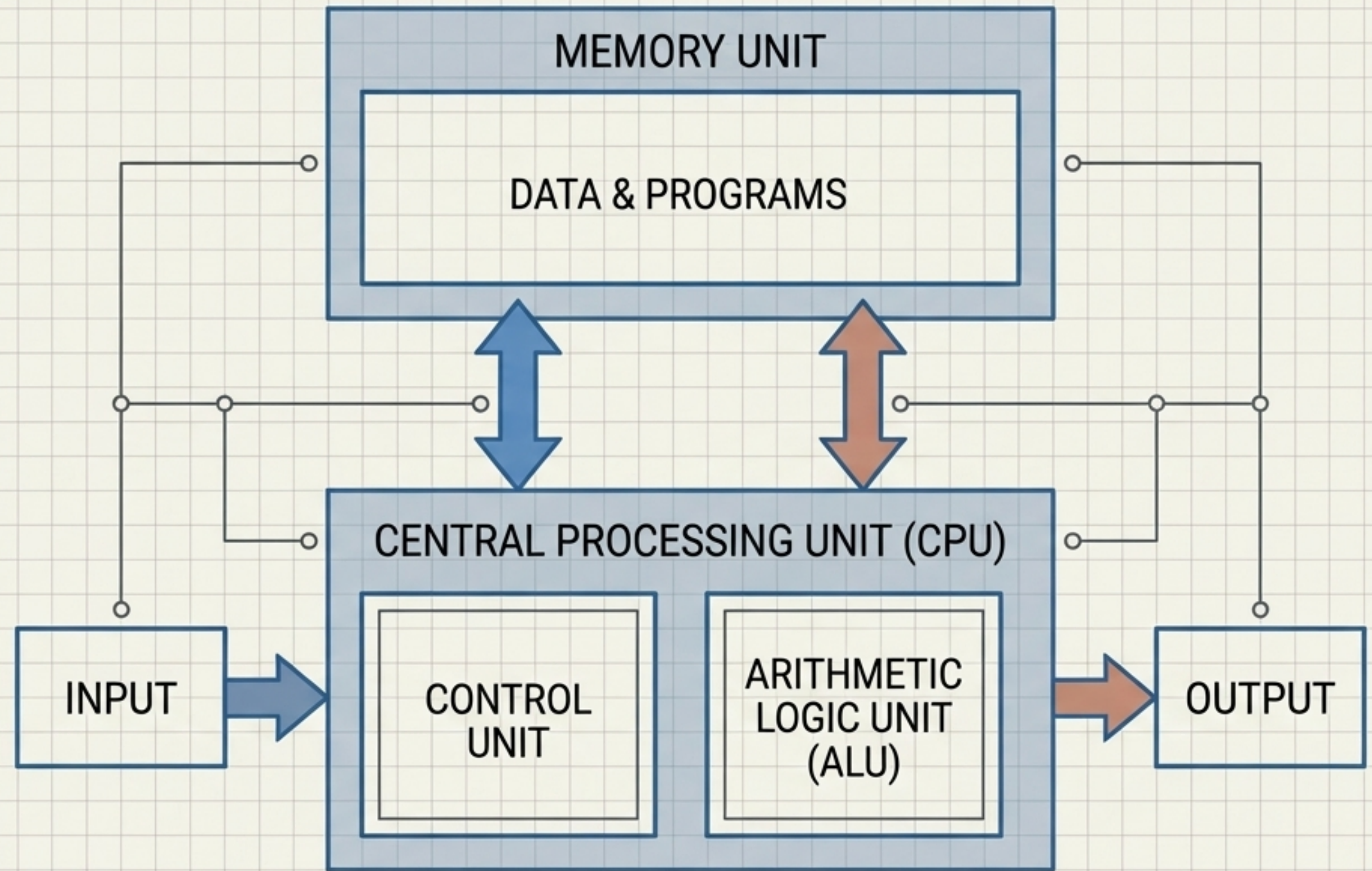
THE ACTION: OPENING A FILE



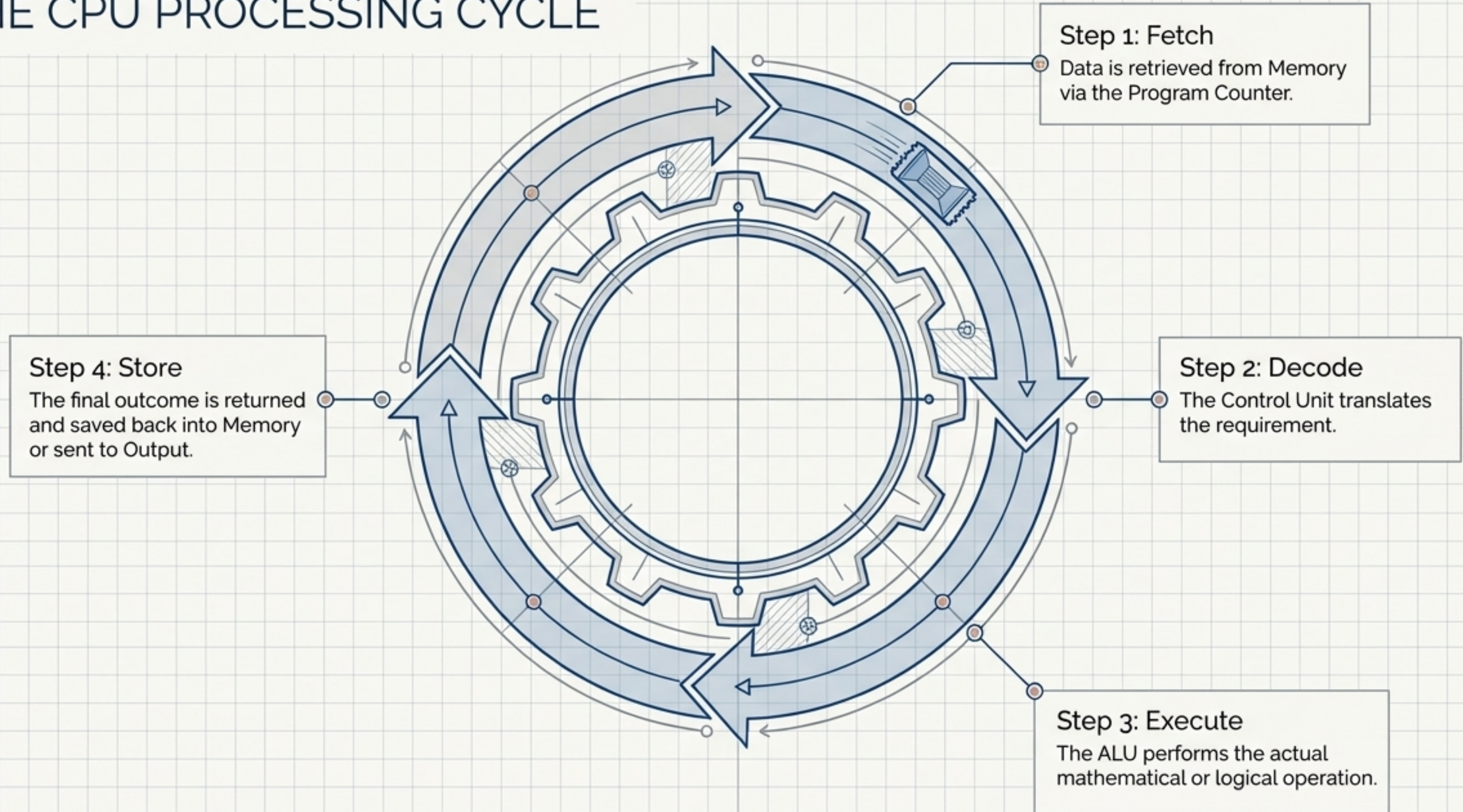
THE VON NEUMANN BLUEPRINT (1940)

Before 1940, systems like the Harvard Architecture separated data and instructions. Mathematician John von Neumann introduced a unified breakthrough that still powers modern laptops today.

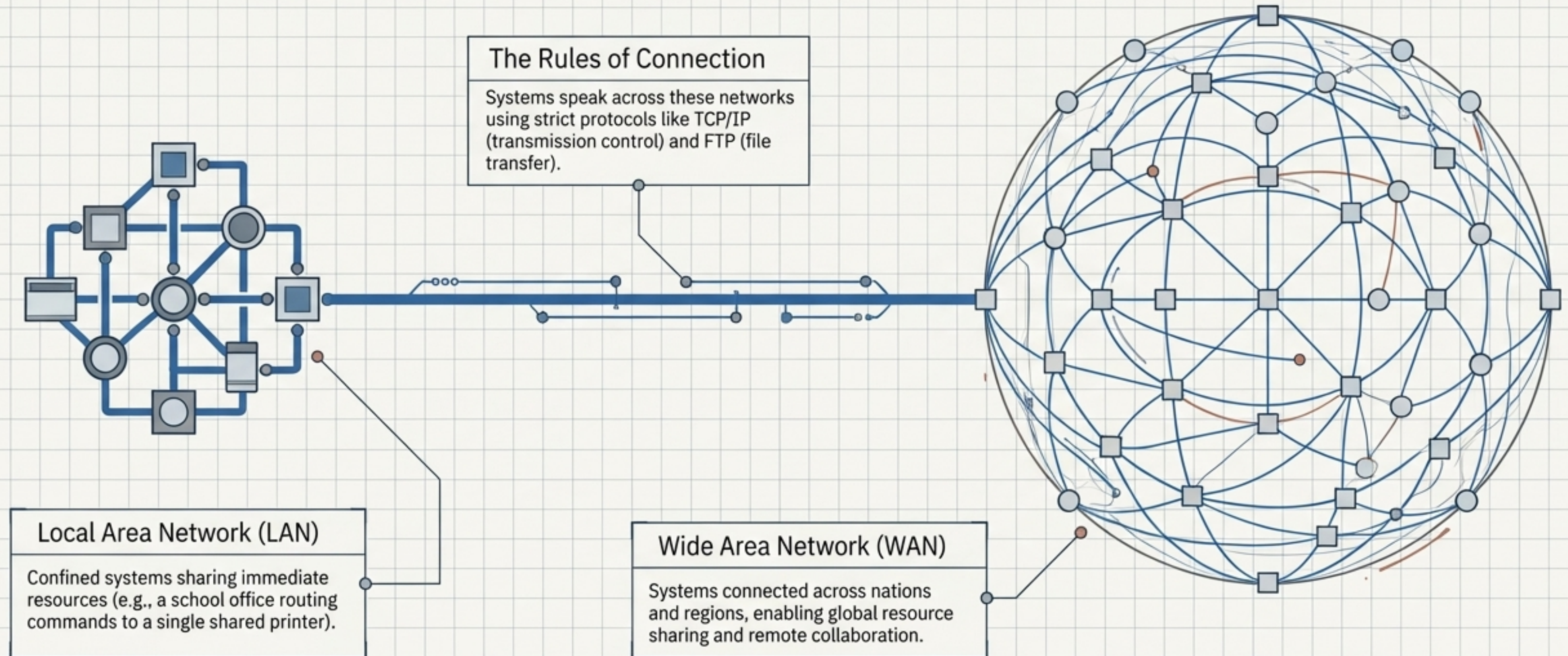
Core Innovation: A single, unified memory unit that holds both the input data and the program instructions, feeding them into a Central Processing Unit (CPU) where the Control Unit acts as the ultimate brain.



THE CPU PROCESSING CYCLE



EXPANDING THE NODE: NETWORK SYSTEMS



THE JOURNEY OF A SYSTEM

